

Computer Networks
CL3001

LAB - 11
OSPF, BGP

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Computer Networks Lab 11

Course: Computer Networks (CL3001)

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Note:

- Maintain discipline during the lab.
- Listen and follow the instructions as they are given.
- Just raise hand if you have any problem.
- Completing all tasks of each lab is compulsory.
- Get your lab checked at the end of the session.

ROUTING PROTOCOL

1. Default Route

We use default routing to send packets with a remote destination network not in the routing table to the next-hop router. You should only use default routing on stub networks—those with only one exit path out of the network.

2. Configuration of Static Route & Default Route

Using Cisco Packet Tracer software, we will now simulate the following network mentioned in the below figure which has 4 networks and 10 subnetworks and assign each host an IP. Also we have to assign an IP for each interface on the router. We assign an IP for the Router interface and start it up using the following commands:

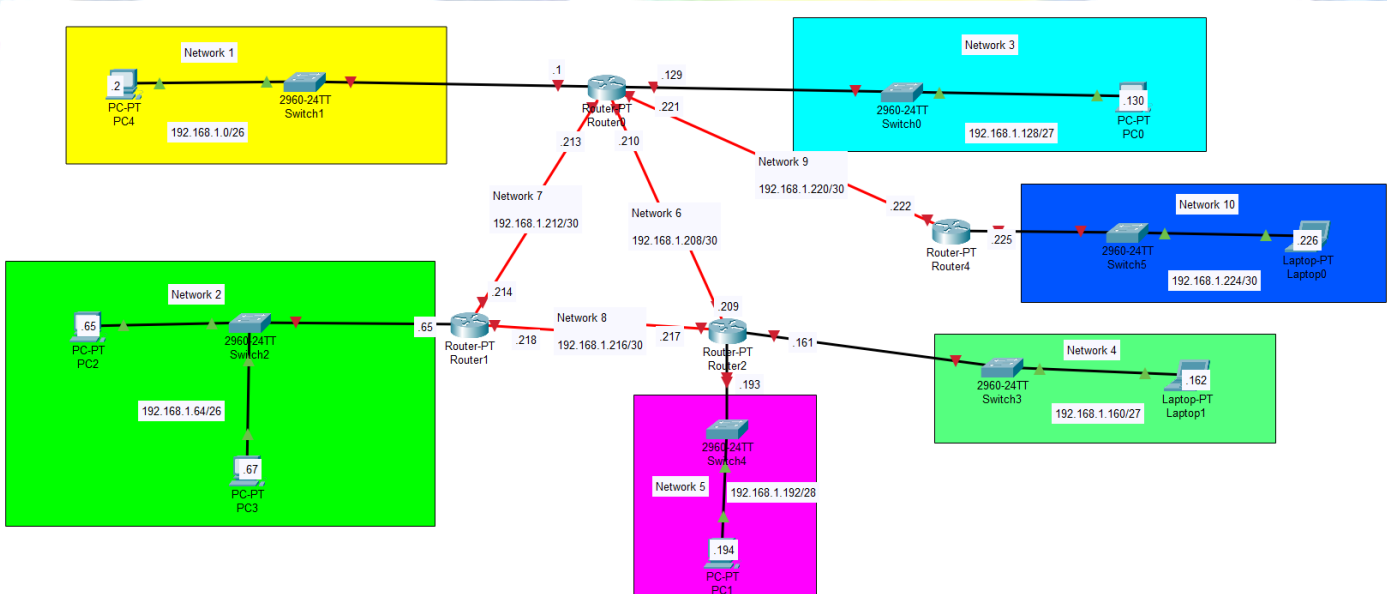


Figure 1

```
Router(config)#interface fa0/0
Router(config-if)#ip address 192.168.1.129 255.255.255.192
Router(config-if)#no shutdown
```

Where:

fa0/0: is the name of the interface.

192.168.1.129: is the IP address for interface fa0/0.

255.255.255.192: is the subnet mask being used on the network that connected directly to the interface.

no shutdown: to start up the interface.

Run the same commands for all routers interfaces and assign each interface an appropriate IP/mask pair.

Now we start routing:

Network 10 is connected directly to Router 3 and no other subnets is connected to Router 3, so we can configure default route on it using the following command:

```
Router(config)#ip route 0.0.0.0 0.0.0.0 192.168.1.221
```

Where:

0.0.0.0: is the destination network IP [0.0.0.0 in case of default routing]

0.0.0.0: is the subnet mask being used on the destination network.

192.168.1.221: is the address of the next-hop router that will receive the packet and forward it to the destination network.

For the other routers, we cannot implement default routing since each of them is connected to more than one network. In this case, we use static routing. We can configure static route on router0 as follow:

```
Router(config)#ip route 192.168.1.64 255.255.255.192 192.168.1.214
```

Where:

192.168.1.64: is the destination network we want to send packets to it.

255.255.255.192: is the subnet mask being used on the destination network.

192.168.1.214: is the address of the next-hop router that will receive the packet and forward it to the destination network.

Configuring all other static routes on router0:

```
Router(config)#ip route 192.168.1.160 255.255.255.224 192.168.1.209
```

```
Router(config)#ip route 192.168.1.192 255.255.255.240 192.168.1.209
```

```
Router(config)#ip route 192.168.1.224 255.255.255.252 192.168.1.222
```


Do the same thing for the other routers. To review routing table on a router, use command “show ip route” as shown in the below figure:

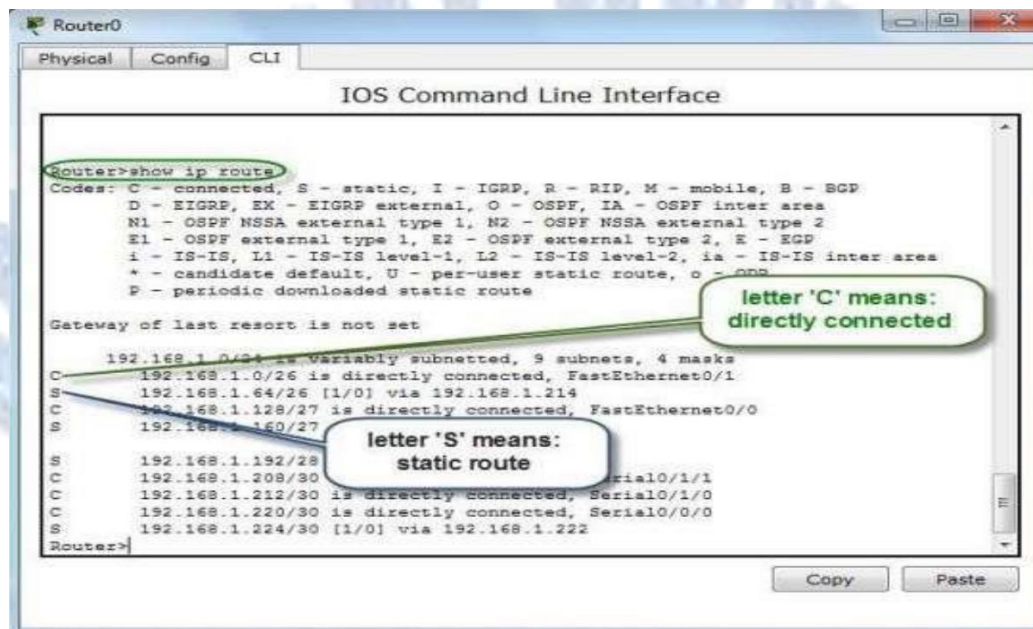


Figure 2

Alternatively, by using the Inspect tool from the right panel, and select “Routing Table” from the menu we can also view the routing table:

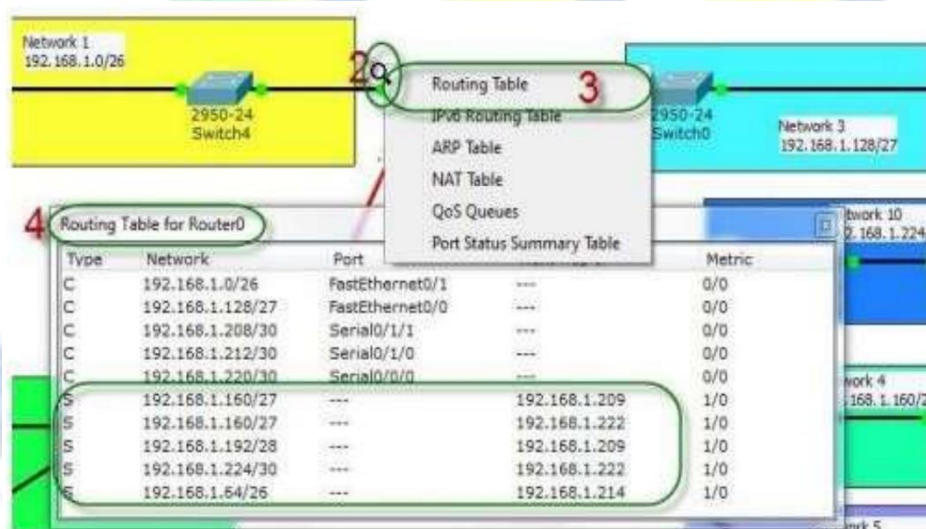


Figure 3

Another quick note: to mention that when (if) the packet is lost on the way back to the originating host, you will typically see a “Request timed out” message because it is an unknown error shown in figure 4. If the error occurs because of a known issue, such as if a route is not in the routing table on the way to the destination device, you will see a “Destination host unreachable” message as shown in the below figure. This should help you determine if the problem occurred on the way to the destination or on the way.

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ping 192.168.1.66

Pinging 192.168.1.66 with 32 bytes of data:

Request timed out.
Request timed out.
Request timed out.
Request timed out.

Ping statistics for 192.168.1.66:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PC>|
```

Figure 4

Destination host unreachable message:

Command Prompt

```
Packet Tracer PC Command Line 1.0
PC>ping 192.168.1.162

Pinging 192.168.1.162 with 32 bytes of data:

Reply from 192.168.1.65: Destination host unreachable.
Reply from 192.168.1.65: Destination host unreachable.
Reply from 192.168.1.65: Destination host unreachable.
Reply from 192.168.1.65: Destination host unreachable.

Ping statistics for 192.168.1.162:
    Packets: Sent = 4, Received = 0, Lost = 4 (100% loss),
PC>|
```

Figure 5

3. Dynamic Routing Types & Their Configuration

Open Shortest Path First (OSPF)

OSPF works by using the Dijkstra algorithm. First, a shortest path tree is constructed, and then the routing table is populated with the resulting best paths. OSPF converges quickly, although perhaps not as quickly as EIGRP, and it supports multiple, equal-cost routes to the same destination. Like EIGRP, it does support both IP and IPv6 routed protocols.

OSPF Terminology:

- Router ID.
- Neighbor.
- Adjacency.
- Hello protocol.
- Neighborship database.
- Link State Advertisement.
- Topological database.
- OSPF areas.
- Loopback address.
- Link costs.

OSPF provides the following features:

- Consists of areas and autonomous systems.
- Minimizes routing update traffic.
- Allows scalability.
- Supports VLSM/CIDR.
- Has unlimited hop count.
- Allows multi-vendor deployment (open standard).

OSPF Practical Part

Lab_10#config t Lab_A(config)#router ospf 1

1: is OSPF process number: out of scope for CCNA, range: <1-65535>

Lab_10(config-router)#network 10.0.0.0 0.255.255.255 area 0 0.255.255.255 : an example of wildcard

Wildcard

The wildcard mask can be configured as the inverse of a subnet mask. For example, IP 172.16.1.16/28 network. The subnet mask for this interface is /28 or 255.255.255.240. The inverse of the subnet mask results in the wildcard mask.


```

255.255.255.255
- 255.255.255.240 (Subtract the subnet mask)

```

```

-----

```

0. 0. 0. 15 Wildcard mask

```

255.255.255.255
255.255.255.0 (Subtract the subnet mask)

```

```

-----

```

0. 0. 0. 255 Wildcard mask

```

255.255.255.255
255.255.0.0 (Subtract the subnet mask)

```

```

-----

```

0. 0. 255. 255 Wildcard mask

Border Gateway Protocol (BGP)

Border Gateway Protocol (BGP) is a true path-vector routing protocol. BGP maintains a table of network paths and makes routing decisions based on path, network policies, and rule sets. It is primarily used for routing between autonomous systems (AS) on the internet. Instead of relying on hop count, BGP considers multiple attributes like AS path, next-hop, and local preference to determine the best route. BGP ensures stable and scalable inter-domain routing but requires careful configuration and management to prevent routing loops and optimize performance.

Below figure shows a topology with BGP:

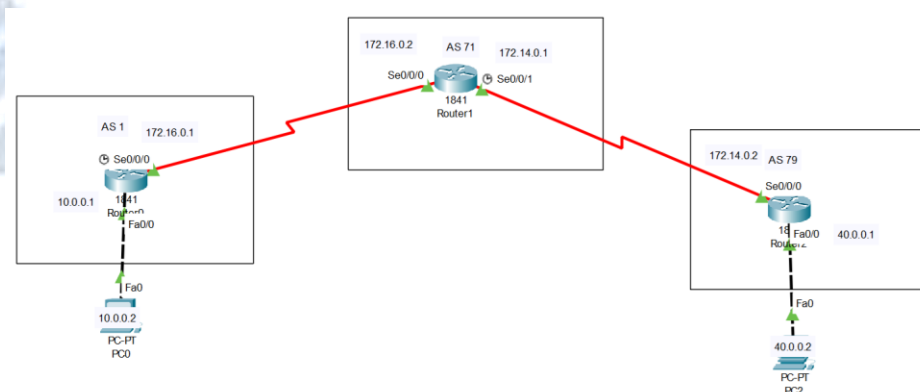


Figure 6

Assign IPs on PCs:

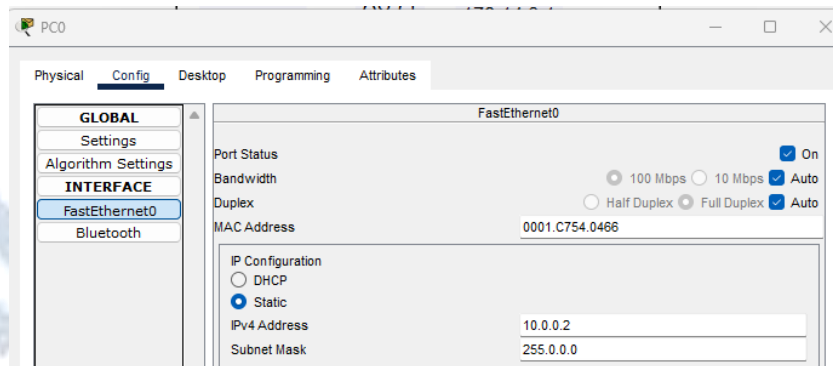


Figure 7

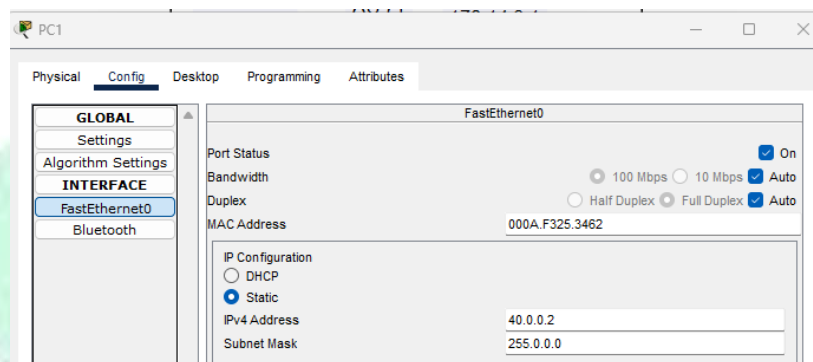


Figure 8

Configuration on Router 0:

```
R0(config)#router bgp 1
R0(config-router)#neighbor 172.16.0.2 remote-as 71
R0(config-router)#network 10.0.0.0 mask 255.0.0.0
R0(config-router)#exit
R0(config)#do write
Building configuration...
[OK]
R0(config)#
```

Configuration on Router 1:

```
R1(config)#router bgp 71
R1(config-router)#neighbor 172.16.0.1 remote-as 1
R1(config-router)#%BGP-5-ADJCHANGE: neighbor 172.16.0.1 Up

R1(config-router)#neighbor 172.14.0.2 remote-as 79
R1(config-router)#network 40.0.0.0 mask 255.0.0.0
R1(config-router)#exit
R1(config)#do write
Building configuration...
```


[OK]

R1(config)#

Configuration on Router 2:

R3(config)#router bgp 79

R3(config-router)#neighbor 172.14.0.1 remote-as 71

R3(config-router)#%BGP-5-ADJCHANGE: neighbor 172.14.0.1 Up

R3(config-router)#network 40.0.0.0 mask 255.0.0.0

R3(config-router)#exit

R3(config)#do write

Building configuration...

[OK]

R3(config)#

BGP Testing & Troubleshooting

For bgp testing we will ping both pc and check the network communication.

On PC1,

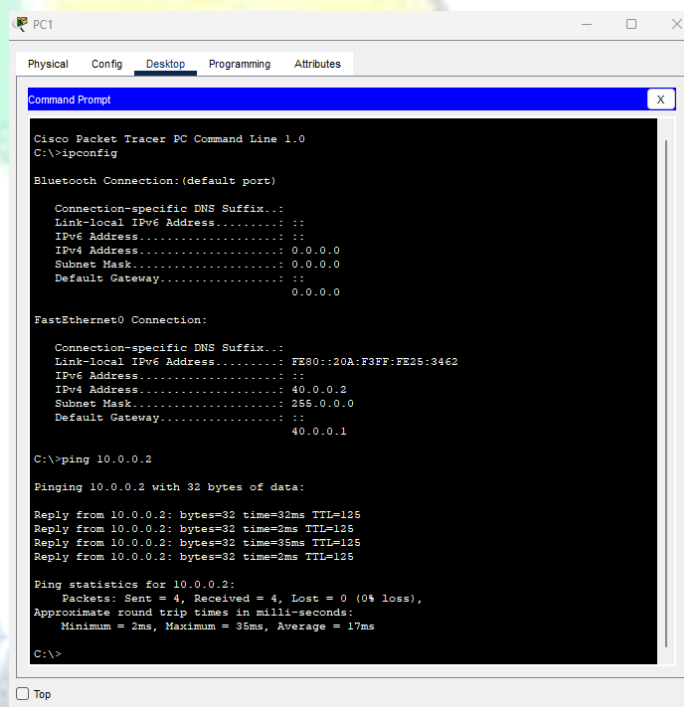


Figure 9

Check BGP route on router R0:

```
R1>enable
R1#
R1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
R1(config)#interface FastEthernet0/1
R1(config-if)#
R1(config-if)#exit
R1(config)#interface FastEthernet0/0
R1(config-if)#exit
R1(config)#
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
        D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
        N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
        E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
        i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
        * - candidate default, U - per-user static route, o - ODR
        P - periodic downloaded static route

Gateway of last resort is not set

C    10.0.0.0/8 is directly connected, FastEthernet0/0
B    40.0.0.0/8 [20/0] via 172.16.0.2, 00:00:00
C    172.16.0.0/16 is directly connected, Serial0/0/0

R1#
```

Check whether BGP protocols configured or not on Router R0:

```
R1>en
R1#show ip protocols
Routing Protocol is "bgp 1"
  Outgoing update filter list for all interfaces is not set
  Incoming update filter list for all interfaces is not set
  IGP synchronization is disabled
  Automatic route summarization is disabled
  Neighbor(s):
    Address          FiltIn FiltOut DistIn DistOut Weight RouteMap
    172.16.0.2
  Maximum path: 1
  Routing Information Sources:
    Gateway         Distance      Last Update
    172.16.0.2             20          00:00:00
  Distance: external 20 internal 200 local 200

R1#
```

Show BGP Status:

```
R1>en
R1#show ip bgp summary
BGP router identifier 172.16.0.1, local AS number 1
BGP table version is 5, main routing table version 6
2 network entries using 264 bytes of memory
2 path entries using 104 bytes of memory
1/1 BGP path/bestpath attribute entries using 184 bytes of memory
2 BGP AS-PATH entries using 48 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 1 (at peak 1) using 32 bytes of memory
BGP using 632 total bytes of memory
BGP activity 2/0 prefixes, 2/0 paths, scan interval 60 secs

Neighbor      V    AS MsgRcvd MsgSent  TblVer  InQ OutQ Up/Down State/PfxRcd
172.16.0.2    4     71    146    140      5     0     0 01:15:45         4

R1#
```

Show BGP neighbor's status:

```
R1>en
R1#show ip bgp neighbors
BGP neighbor is 172.16.0.2, remote AS 71, external link
  BGP version 4, remote router ID 172.16.0.2
  BGP state = Established, up for 01:16:49
  Last read 01:16:49, last write 01:16:49, hold time is 180, keepalive interval is 60
seconds
  Neighbor capabilities:
    Route refresh: advertised and received(new)
    Address family IPv4 Unicast: advertised and received
  Message statistics:
    InQ depth is 0
    OutQ depth is 0

      Sent      Rcvd
Opens:          2         2
Notifications:  0         1
Updates:        2         4
Keepalives:    139       140
Route Refresh:  0         2
Total:         143       149
Default minimum time between advertisements runs is 30 seconds

For address family: IPv4 Unicast
BGP table version 5, neighbor version 6/0
Output queue size : 0
Index 1, Offset 0, Mask 0x2
1 update-group member

      Sent      Rcvd
Prefix activity:
-----
Prefixes Current:    1         1 (Consumes 46 bytes)
Prefixes total:      1         1
Implicit Withdraw:    0         0
Explicit Withdraw:    0         0
Used as bestpath:    n/a        1
Used as multipath:    n/a        0
```

Similarly, we can check the BGP route on every other Routers as well by verifying the advertised and received routes. This helps ensure proper path selection and confirms that routing policies are applied correctly.